

Dataset and benchmark for detecting moving objects in construction sites

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ABSTRACT

Detecting workers and equipment through images/videos can assist in safety monitoring, quality control, and productivity management at construction sites. Currently, the dominant method for detecting is Deep Neural Networks (DNNs). To apply this method, the DNNs always need to be trained on image datasets that contain objects at the construction site. However, a large-scale and publicly available image dataset for detecting objects at construction sites is still absent, and this hinders research in this field. In this study, the Moving Objects in Construction Sites (MOCS) image dataset is presented. The dataset contains 41,668 images collected from 174 different construction sites. Thirteen categories of moving objects found in construction sites were annotated. Furthermore, the objects were precisely annotated using per-pixel segmentations to assist in precise object localization. A detailed statistical analysis was performed in this study. Finally, a benchmark containing 15 different DNN-based detectors was made using the MOCS dataset. The results show that all detectors trained on the dataset could detect objects at construction sites precisely and robustly.

1. Introduction

The automatic process of construction management using computer vision techniques has aroused the interest of both academia and industry. These techniques only require digital video and images of the construction site, which are easily accessible by inexpensive equipment, such as digital cameras, surveillance cameras, smart phones, and unmanned aerial vehicles (UAVs). Numerous applications have been developed, such as productivity analysis [1], progress monitoring [2], and safety monitoring [3]. For these applications, accurate detection/segmentation and correct identification of moving objects (e.g. workers, loaders, excavators, cranes, etc.) through digital images in construction sites are one of the basic requirement [1–5]. However, the detectors employed in previous studies [1–3] are mainly based on hand-crafted features, and this greatly limits the generalization of applications [4].

Recently, deep neural networks (DNNs) have demonstrated superior performance in object detection. Compared with previous artificially designed detectors, the DNN-based detectors can learn fine-grained features from a large set of data and demonstrate more-robust operation in object detection [4,6–8]. At the same time, large-scale and publicly available datasets (e.g., Caltech [9,10], COCO [11], and Pascal

Visual Object Classes (Pascal VOC) [12,13]) make it convenient to develop, apply, and fairly compare these detectors, and they continuously push the research to state-of-the-art performance [9–13].

DNNs are also leading to a revolution in the construction field. Numerous applications have been developed, such as struck-by hazard prevention [4], earth-moving equipment analysis [5], hardhat-wearing detection [14], noncertified work detection [15], and recognition of construction activities [16].

However, the detectors of the above studies are trained and tested on their own datasets, which could be small-scale (i.e., only a few thousand images) and customized (i.e., targeted at specific objects, or the images are obtained from limited sites, devices, viewpoints, or weather and illumination conditions). As a result, detectors trained with a limited source of data would involve high variance and be difficult to generalize to the usual construction environment [4,5]. Consequently, a large-scale dataset with strong diversity for construction object detection must be developed and made publicly available.

Previous studies only detected coarse locations of objects (i.e., using a bounding box to present the location of objects in images). However, recently developed detectors, such as Mask-RCNN [17], can already detect the location of objects with high precision (i.e., pixel-level

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Fig. 1. Object detection (top) and instance segmentation (bottom).

segmentation masks). This can be useful for understanding the layouts, poses, and activities of objects. Therefore, a dataset with precise annotation is needed for the construction industry to achieve state-of-the-art performance and spawn more applications.

Furthermore, DNN-based detectors are emerging rapidly, and different detectors focus on different detection performances (e.g., accuracy, speed, and small scale objects). The performance of these detectors on common datasets, such as COCO [11] and Pascal VOC [12,13], has been reported. However, construction objects in images are different from common objects. This is because of the shooting distance, illumination conditions, and occlusions. The discrepancy between common datasets and construction datasets could lead to different detector performance on the two datasets. Therefore, a benchmark for DNN-based detectors on a large-scale construction dataset is required.

To address the aforementioned issues, a large-scale dataset called “Moving Objects in Construction Site” (MOCS) is presented. The MOCS dataset contains 41,668 images collected from 174 construction sites. Images of datasets are collected by five different devices and various viewpoints, illumination conditions, and weather conditions. Meanwhile, the dataset is precisely annotated with 13 categories of moving objects in the construction site by the styles of both bounding boxes and masks. There are 222,861 annotated instances in the dataset, and the training and validation dataset with the annotation is made publicly available at http://www.anlab340.com/Archives/IndexArctype/index/t_id/17.html. Furthermore, a benchmark containing 15 different DNN-based detectors was created on the dataset. Test results show that detectors trained on the MOCS dataset could precisely detect moving objects in construction sites. It is believed that the dataset can make it convenient for researchers and engineers to train, test, and develop detection algorithms and computer-vision-based applications. The benchmark in this study can provide a guideline for others to select detectors.

2. Related works

2.1. Object detection

Object detection involves developing computational models and techniques that provide one of the most basic pieces of information needed by computer vision applications — knowledge of what objects are where [18]. Therefore, the task always contains two parallel sub-tasks: localizing objects in an image and classifying the categories of

objects [11]. The localizing task in object detection is to mark out the bounding boxes around each object and the classification task is to classify categories of each object in the boxes [19]. Early researchers tried to solve the problem by the idea of “recognition by components”: using the measurement of similarity between the object components, shapes, and contours to detect objects [18]. Despite the promising initial results, the proposed methods did not work well on more-complicated detection problems until machine-learning algorithms were proposed [18]. From 2004 to 2014, multiple machine-learning-based detectors were developed, such as a histogram-of-oriented-gradients detector [20], deformable-part-based model [21], and the OverFeat detector [22]. With the emergence of deep learning and large-scale datasets, object detection started to evolve at an unprecedented speed after 2014.

In the field of construction, detecting resources in visual data has been a common research focus that has yielded promising results [5]. Roberts et al. trained a crane detector on 456 images to monitor crane-related safety hazards [23]. Kim et al. trained YOLO-v3 on 4512 frames to detect workers and heavy machines and further evaluated the proximity between them to prevent struck-by hazards [4]. Roberts et al. trained a DNN-based detector on an advanced infrastructure management dataset (AIM) [24], which contains 2920 images of five different heavy machines used in construction projects. The trained detectors were further used to analyze the productivity of excavators and trucks [5]. A common issue raised by all the above studies is that, to train a detector for these tasks, a dataset needs to be collected and annotated by researchers, which could cost considerable time and labor. Meanwhile, the collected datasets are often small scale and limited in specific scenarios (e.g., single project, single viewpoint, and limited objects), greatly reducing the generalization performance of the algorithm.

2.2. Instance segmentation

As the accuracy of object detection improves rapidly, a more challenging task has aroused the interest of the computer vision community: instance segmentation [19]. Unlike object detection, instance segmentation must detect all instances of each category and, for each instance, correctly mark all pixels that belong to it [19] (Fig. 1). The problem was first raised in a previous report [19] and has been extensively studied in recent years. Multiple methods have been proposed. Zhang et al. attempted to predict instance labels with a Markov random field (MRF) model [25]. Armb and Torr used a conditional random field to identify instances [26]. In 2017, He et al. added an extra branch based on Faster



Fig. 2. Different types of construction project in MOCS dataset.

R-CNN [8] to obtain pixel-level mask predictions and yield promising results [17].

In the construction field, instance segmentation is also used to better understand the construction scene. Balali and Golparvar employed an MRF model to segment images and recognize roadway assets [27]. Atkinson et al. used a DNN-based detector to identify floorboards, joists, air vents, and pipework [28]. Wei et al. also used a DNN-based detector to localize the bughole of a concrete surface [29]. A brief review of instance segmentation reveals that, although instance segmentation could provide a more accurate location and layout of moving objects in construction sites, few studies and datasets have been proposed for segmentation of moving objects in construction sites.

2.3. Public datasets for object detection and instance segmentation

2.3.1. Datasets for common objects

In the past 10 years, a number of well-known datasets and benchmarks have been released. This has greatly promoted the development of both academia and industry. For example, the Pascal Visual Object Classes (Pascal VOC)¹ challenge is one of the most important competitions in the early computer vision community [12,13]. Two versions are mostly used in object detection: VOC07, which contains 5000 images with 12,000 annotated objects, and VOC12, which contains 11,000 images and 27,000 annotated objects. Twenty classes of objects that are common are annotated in these two datasets, such as a person, dog, bottle, or chair. For details of the dataset, readers are referred to a previous article [13]. Currently, the COCO² dataset is considered the most challenging object detection dataset [11,18]. For example, COCO2017 contains 164,000 images and 897,000 annotated objects of 80 categories. Compared with Pascal VOC, one of the biggest advances in the COCO2017 dataset is that each object is further annotated with

per-instance segmentation to facilitate precise instance segmentation. In addition, COCO2017 contains more small objects (area smaller than 1% of the image) and more densely located objects than the Pascal VOC. There are several other datasets as well, such as Caltech-101 [11] (9146 images of 101 categories) and Caltech-256 [40] (30,607 images of 256 categories), which are used extensively. Hereinafter, “Pascal VOC” refers to Pascal VOC2012 and “COCO” refers to COCO2017.

2.3.2. Datasets for construction objects

There are few image datasets for detecting construction objects [5]. Roberts and Golparvar-Fard released a dataset for detecting, tracking and analyzing earthmoving equipment [5]. The dataset contained 479 atomic activity videos for excavators and dump trucks. Two objects are annotated in each frame, and the videos are annotated by the activity type of equipment, such as “Loading Bucket,” “Swinging Bucket,” and “Moving.” Wu et al. released the GDUT-HWD dataset for hardhat-wearing detection [30]. The dataset contains 3174 images, and 18,893 hardhats are annotated with the type of bounding box. Kim et al. created a dataset that contains 2920 instances of five different categories [24]. The images are collected from ImageNet, which is a large-scale dataset for object classification.

An overview of related works shows that a large-scale image dataset for objects in construction sites is urgently needed for both the research community and industry. Various detectors have been used in previous studies. A benchmark should be created to compare the performance of the detectors impartially.

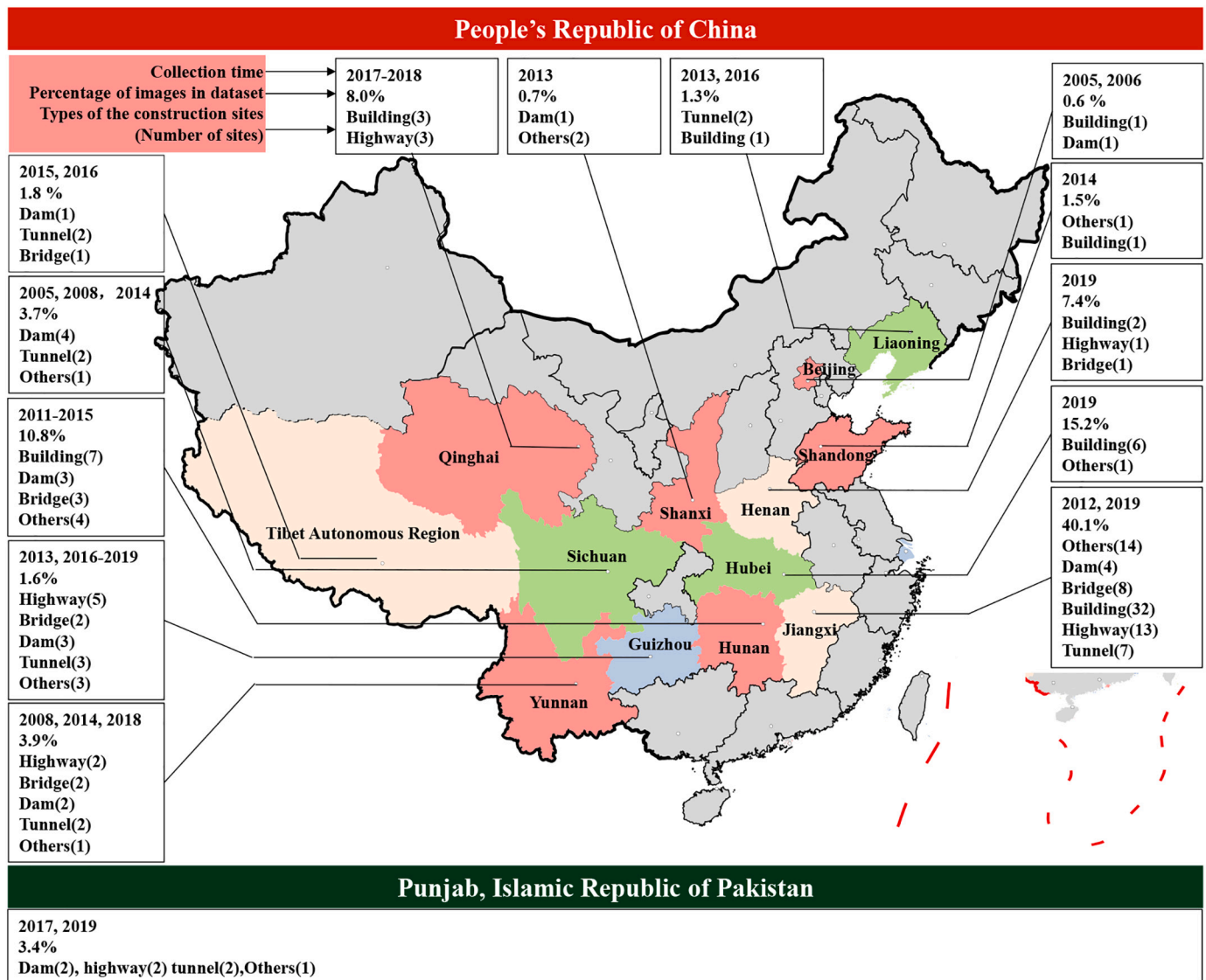
3. Image collection and preparation

A large-scale dataset was constructed in this study. Images in the dataset were collected from 174 different construction project sites. The type of construction site, devices used to collect images, and diversity of the dataset are illustrated in this section.

Common datasets always collect images from web engines, such as Google, Bing, and Baidu [16]. However, images from the web are always

¹ <http://host.robots.ox.ac.uk/pascal/VOC/>

² <http://cocodataset.org/>



This map does not represent any political views and is only used to represent geographic information.

Fig. 3. Origin information of the images in MOCS dataset.

photographed in an iconic view [11]. For example, when performing a web-based image search for the category “Loader,” the top-ranked retrieved examples appear in profile, unobstructed near the center of a neatly composed photo. However, construction sites are chaotic and disordered. Objects in the site images are dense and mixed in the background. Meanwhile, it is difficult to obtain the best distance and viewpoint when capturing images at construction sites. As a result, objects could be extremely small in images, partially occluded, and cluttered. Therefore, to keep the dataset consistent with the construction site, images in the MOCS dataset were collected from construction sites instead of the Internet.

To collect images, 174 construction sites were visited. These sites are sites of building, highway, bridge, tunnel, dam, and other (e.g., sidewalks and interior decoration) projects in China and Pakistan from 2005 to 2019 (Fig. 2). Because of the privacy requirements of the construction company, the specific project name is omitted. As an alternative, the year the images were collected, the location of the images collected, and the percentage of the region’s images in the dataset are shown in Fig. 3. Meanwhile, for each type of construction project, images of MOCS covered different types of specific works (e.g., earth work, concrete

work, form work, scaffold work, steel work, decoration, and assembling) (Fig. 4).

Previous research has employed different devices to acquire data from construction sites. For example, Kim et al. employed a UAV to monitor the proximity of objects [4]. Fang et al. used a surveillance camera to monitor the failure to use a hardhat [12]. Luo et al. collected 763 images from a construction site using digital cameras [14]. To satisfy different requirements, images of the MOCS dataset were collected by various devices (Fig. 5). In the dataset, 52.7% of the images were collected by the authors. Four imaging devices (a smartphone³, a UAV⁴, and two digital cameras⁵) were employed to collect images from the construction site. For the UAV, a distance of 10–20 m from the ground was maintained when collecting images. Meanwhile, the UAV was set to circle around the site to collect images from different views.

³ Two types of smartphone were used: iPhone 8 plus and HUAWEI P30.

⁴ The UAV used in this research was Mavic Pro from Da-Jiang Innovations.

⁵ The two digital cameras used in this research were GoPro HERO Session and Canon EOS 6D.



Fig. 4. Different types of construction work in MOCS dataset.



Fig. 5. Devices used to collect images.

For the smartphone and digital cameras, the sites were visited, and images of various angles and distances were randomly collected. Of the images, 47.3% were provided by related construction companies. Some

of these images were extracted from on-site monitoring cameras, and some were images recorded by site managers using mobile phones or digital cameras. Moreover, images were collected under different



Fig. 6. Samples of different poses, viewpoints, illumination, weather conditions, and occlusions in MOCS.

weather conditions, such as sunny, foggy, rainy, and snowy conditions. Meanwhile, to keep the dataset consistent with the construction sites, no additional illumination equipment was used. Most images were collected under sunlight. For projects that need to be conducted at night or in tunnels, images were collected under the illumination conditions at the site. Finally, images of objects were not deliberately collected from a specific viewpoint; instead, the sites' environments were recorded. Therefore, objects were often occluded in the images. Such strategies can be used to capture images with different viewpoints, weather conditions, and illumination conditions. Meanwhile, objects in images present various poses and are sometimes occluded by other objects (Fig. 6).

In summary, 43,000 images were collected from 174 different construction sites. Images were captured by five different imaging devices with different viewpoints, distances, and resolutions. At the same time, the MOCS images cover different types of construction project and different types of construction work. However, because different kinds of high-resolution camera were used, the encodings and the sizes of images were different, and the storage space of the dataset was too large. Therefore, all images were converted to a red–green–blue color space and stored in JPG format. Then, all images were resized using OpenCV [57], and the interpolation method used was interpolation by area relation: the long side of the image was reduced to 1200 pixels, while the scale of the image was kept constant.

4. Image annotation

After image collection, the objects in the images were annotated. The MOCS dataset is designed to train detectors for moving objects in

Table 1
Categories in MOCS dataset and related studies.

Categories	Objective of detection
Worker	Struck-by hazard prevention [4], hard hat detection [14], noncertified work detection [15], activity recognizing [16]
Tower crane	Safety monitoring [23], construction activity recognition [16]
Hanging hook	Safety monitoring [23,31,32]
Truck crane	Construction activity recognition [16], equipment detection [33]
Roller	Construction activity recognition [34], equipment recognizing [35]
Bulldozer	Construction activity recognition [16], equipment detection [33]
Excavator	Construction activity recognition [16,34,36], productivity analysis [5], struck-by hazard prevention [4]
Truck	Construction activity recognition [5,16], equipment detection [33], equipment recognition [35]
Loader	Struck-by hazard prevention [3], equipment detection [33]
Pump truck	Construction activity recognition [16], equipment detection [33]
Concrete truck	Construction activity recognition [16], equipment detection [33]
Pile driver	Position detection [37], pile driver navigation [38]
Other vehicle	Construction activity recognizing [16], equipment detection [33]

construction sites. Therefore, other objects, such as construction materials, structure components, and other construction-unrelated objects, are not annotated in this dataset. To decide which objects needed to be annotated, four construction engineers with 10–20 years of experience were asked. They listed the 12 most common machines the construction sites: tower crane, truck crane, truck, roller, bulldozer, excavator, loader, pump truck, concrete truck, pile driver, road paver, and tunnel boring machine. In the dataset, there are fewer images of road pavers

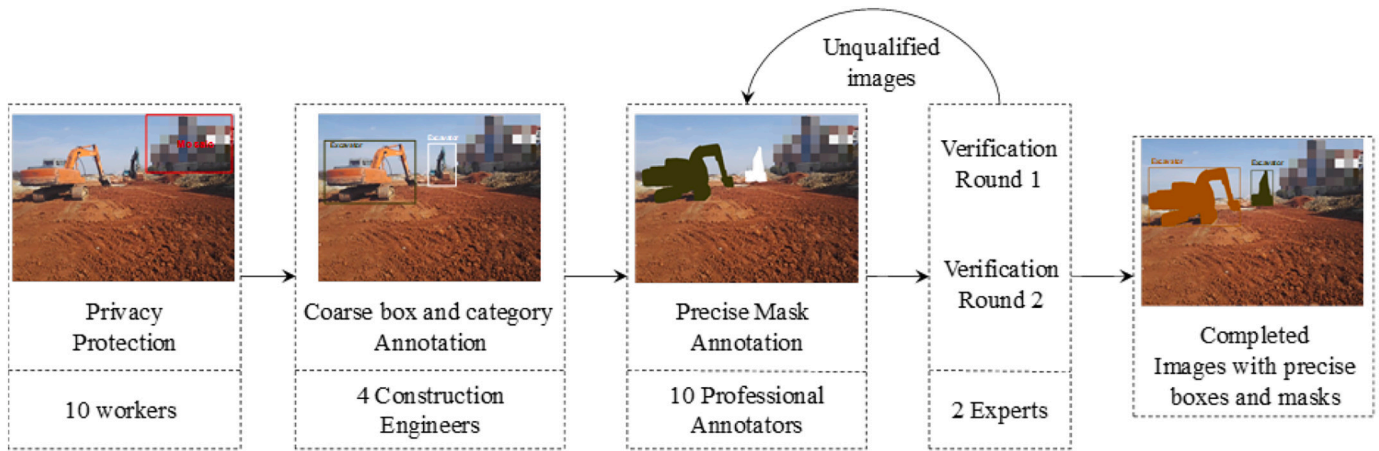


Fig. 7. Illustration of workflow of annotation.

and tunnel boring machines (<1000 images) compared with other machines. Therefore, these two machines were not annotated in the dataset. In addition to the remaining 10 construction-related machines, three other objects were also annotated in the dataset: the first object is the worker, who is the most common object at the construction site. Second, civil vehicles, which are unrelated to construction activities but often appear at a site, were also annotated in the dataset. Detecting these vehicles enables managers to better plan site traffic. Third, the hanging hook of a tower/truck crane is considered a major hazard source in construction sites by many researchers [16,23]. In total, 13 types of moving object were annotated in the dataset. The categories and relevant existing studies are listed in Table 1.

The workflow of the annotation is illustrated in Fig. 7. First, to protect the privacy of individuals and companies, all slogans and biometric information in images were carefully masked with the type of mosaic. This was conducted by 10 workers. Second, the masked images were sent to four experts who were construction engineers and familiar with construction objects. Coarse bounding boxes and categories of corresponding boxes were annotated in the images. Third, 10 annotators, who were professional annotators of an annotation company, were employed to draw the outlines of objects carefully. Then, each annotated image went through two rounds of inspection, and unqualified images were filtered out to revise until all the objects were meticulously annotated. The rejection rate was used to evaluate the quality of the annotation. The rejection rates of the first and second rounds of inspection were 2.626% and 0.055%, respectively.

Finally, 222,861 instances of 13 categories were precisely annotated with the type of bounding box and mask. The final bounding boxes were generated by searching the boundaries of the masks instead of the coarse boxes annotated by four experts. The annotation style of the MOCS dataset is similar to that of the COCO dataset. For each split dataset (i.e., training set, validation set, and test set), there is only one JSON file to store the annotation information. Details of the annotation are illustrated in Fig. 8. All images were shuffled randomly and then encoded from 1 to 41,668. Image examples of each category are shown in Fig. 9 (the color of the mask was generated randomly).

5. Dataset statistics

Next, the properties of MOCS were analyzed and compared with those of several other popular datasets. Because the previous datasets for construction object detection were small, they were not included for comparison. Instead, common datasets that are popular and large scale were compared with MOCS in this study. These datasets are VOC, Caltech-101, Caltech-256, and COCO. These datasets varied

significantly in size, list of categories, and types of image. Pascal VOC, Caltech-101, and Caltech-256 concentrate on object detection in natural images. COCO is designed for detection and segmentation of objects occurring in their natural context. MOCS aims at the detection and segmentation of objects in a construction site context with different viewpoints, illumination conditions, and weather conditions.

In Fig. 10, the centers of objects in MOCS are plotted by the style of the thermograph, which can provide a qualitative understanding of the object distribution in the images. The coordinates of each subplot in Fig. 10 are the normalized image sizes (1:1). The horizontal (transverse) and vertical (longitudinal) axes represent the normalized width and height of the images, respectively. The curves on the top and right of the thermograph represent the distribution of the instance centers in the transverse and longitudinal directions, respectively. For example, most excavators appear in the center of the images. Compared with the excavators, the hanging hook appears more dispersed.

Fig. 11 shows the statistics of the MOCS dataset. The numbers of instances and images for all 13 categories in MOCS are shown in Fig. 11 (a). Each category has more than 1000 images, and “worker” has the most instances and images. This is consistent with practical construction sites. The MOCS dataset was roughly split into three parts for training (19,404 images), validation (4000 images), and testing (18,264 images — approximately 1/2 training, 1/10 validation, and 1/2 testing). The number of instances per category of split datasets is shown in Fig. 11(b). Fig. 11(c) shows the percentage of images with a certain number of instances. For example, images with two instances account for the most images in the MOCS dataset. This is similar to the COCO dataset. The number in the legend represents the average number of instances per image. Fig. 11(d) shows the percentage of images with a certain number of categories, and the number in the legend is the average number of categories per image. Fig. 11(e) shows the statistic of instance size. The horizontal axis is the area proportion of instances in the images, and the vertical axis is the percentage of instances. Similar to COCO, MOCS has more small objects than Pascal VOC. This is induced by the long shooting distance. Finally, the scale of the MOCS dataset is compared with popular datasets, including Caltech-101, Caltech-256, Pascal VOC, and COCO. The results are shown in Fig. 11(f).

6. Benchmarking

Fifteen deep-learning-based detectors were selected and evaluated. Owing to space constraints, only a brief introduction of the evaluation metrics and detectors is provided. The benchmark presented in this article was used to verify whether the detectors trained on the MOCS dataset could detect objects well and to provide a reference for

JSON File

<pre>{ "images": [{ "height": 1200, "width": 900, "id": 2, "file_name": "0019407.jpg" }, ], "categories": [{ "id": 1, "name": "Worker" }, ], "annotations": [{ "segmentation": [[530.0, 454.0, ], [255.0, 450.0, ]], "image_id": 2, "bbox": [226.0, 446.0, 343.0, 50.0], "area": 17150.0, "category_id": 8, "id": 5 }, ] }</pre>	Images For train set, 19404 elements
<pre> { "id": 1, "name": "Worker" }, ],</pre>	Categories 13 elements for 13 categories
<pre> { "segmentation": [[530.0, 454.0, ], [255.0, 450.0, ]], "image_id": 2, "bbox": [226.0, 446.0, 343.0, 50.0], "area": 17150.0, "category_id": 8, "id": 5 }, ]</pre>	Annotations Each element corresponds to one instance Edge points of mask: [[x,y,x,y,x,y.....], (part 1 of this object) [x,y,x,y,x,y.....], (part 2 of this object)] Bounding box annotation: [x,y,width,height] Mask area Instance id

Fig. 8. Example of annotation file of MOCS dataset.

researchers and engineers to select their detectors. Another report focusing on the performance of more detectors is planned. To evaluate the performance of algorithms with less bias, the dataset was split into a training set (19,404 images), validation set (4000 images), and test set (18,264 images) randomly. Fifteen state-of-the-art deep-learning algorithms were selected to train and test the dataset.

6.1. Evaluation metrics

To evaluate the performance of DNN-based detectors on the MOCS dataset, the metrics of the MS-COCO dataset, which has been the de facto metric for object detection for years, were used [18].

As mentioned in Section 2.1, object detection consists of two sub-

tasks: object location and object classification. For the task of object location, a common criterion is the intersection over union (IoU) of the predicted bounding box with the ground truth bounding box. Only if the IoU is above a certain threshold (typically 0.5) is the predicted box considered a valid box. Meanwhile, for the task of classification, the detectors output the category with the highest confidence score for each box. Another threshold for the confidence score is used to delete a large number of boxes with low scores. A detection is treated as a true positive (TP) only if the box is valid and the category of the box is the same as the ground truth. If at least one of these two conditions is not met, the detection is considered a false positive (FP). False negatives (FNs) are ground truth bounding boxes with no sufficiently overlapping detection bounding boxes, as determined by the aforementioned threshold. Two

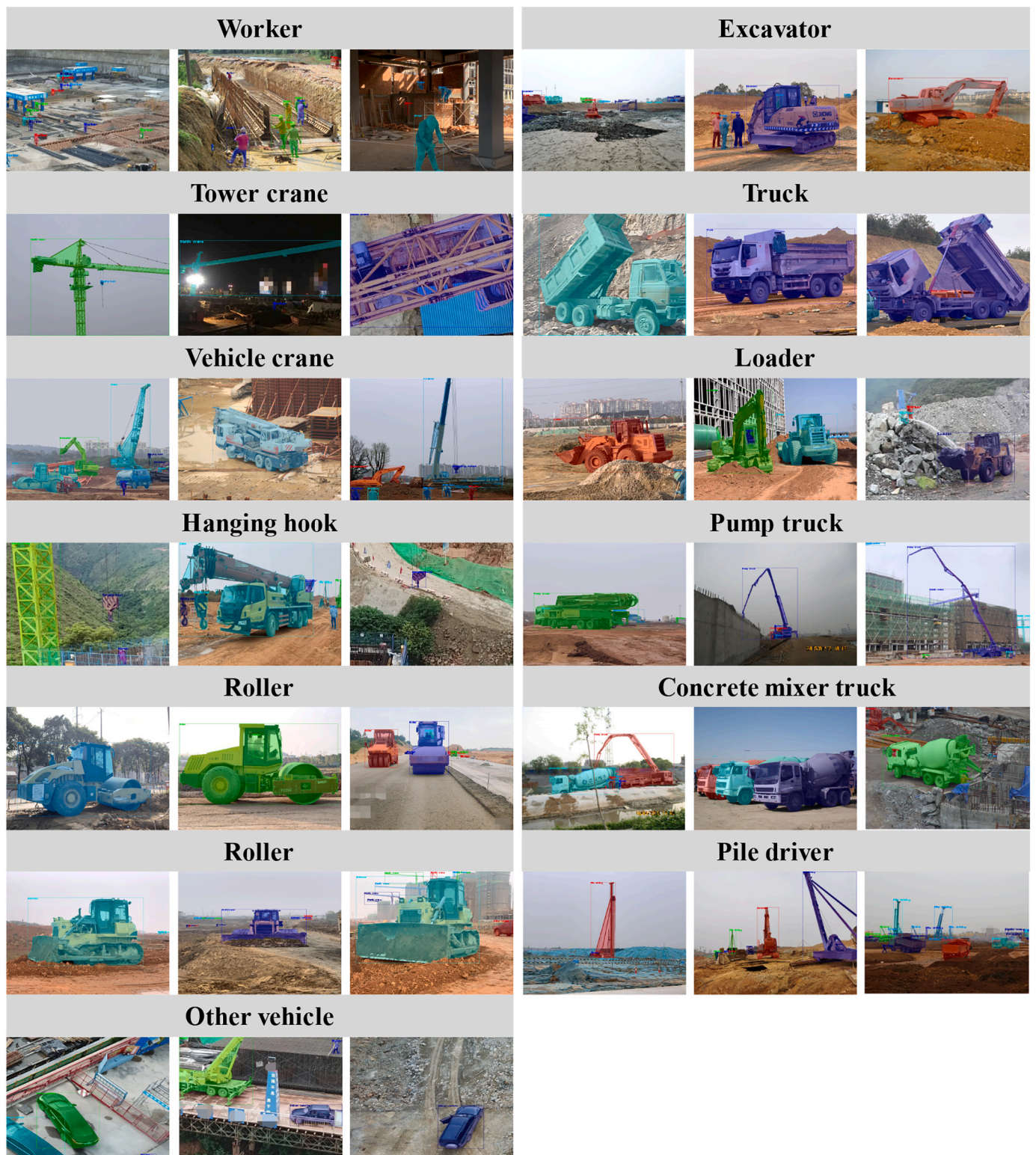


Fig. 9. Image examples of annotated categories in MOCS.

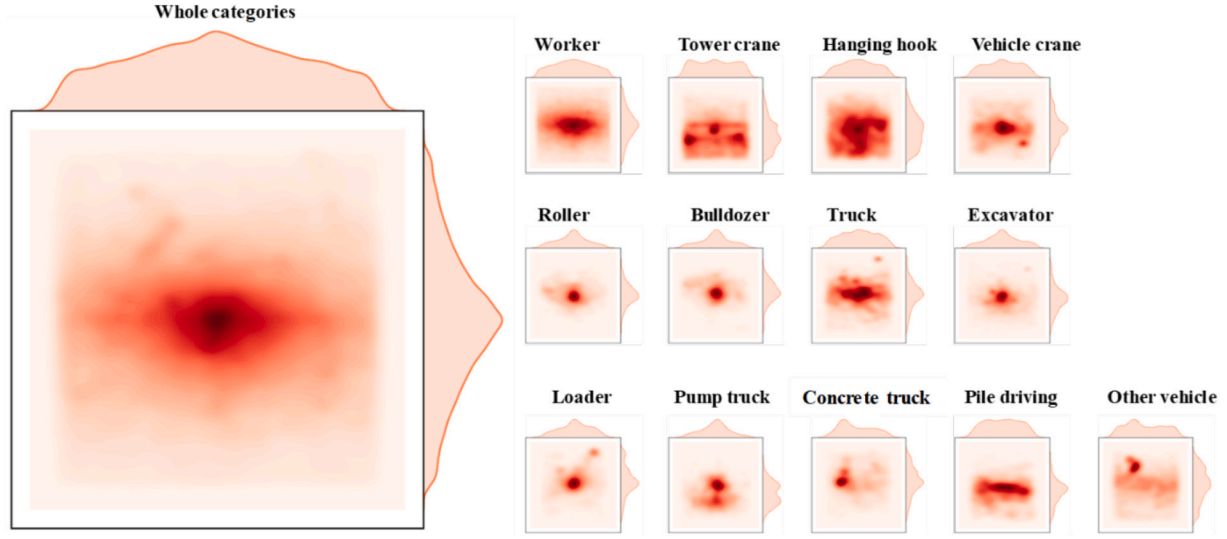


Fig. 10. Visualization of distribution of objects in MOCS.

metrics can be calculated using TP, FP, and FN, as in Eqs. (1) and (2).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

“Precision” denotes the portion of correct boxes in all predicted boxes, while “recall” denotes the portion of correct boxes in all ground truth boxes. The two thresholds mentioned above affect the value of precision and recall.

The first is the threshold of the confidence score. With a high threshold of confidence score, the precision is high, while the recall is low. However, the precision is low and the recall is high with a low threshold. To evaluate fully the detectors under various thresholds, a typical method is to calculate the average precision over varying levels of recall, as in Eq. (3).

$$AP^{101} = \frac{1}{101} \sum_{r \in \{0.01, \dots, 1\}} P_r \quad (3)$$

where AP^{101} is the average precision of 101 levels of recall, and P_r represents the precision at recall level r .

The second is the threshold of IoU. With a high threshold of IoU, both precision and recall are low. In contrast, the two metrics are high. The method to evaluate the performance of detectors on different levels of IoU is to calculate the average AP over multiple IoUs, as follows.

$$AP = \frac{1}{10} \sum_{IoU \in \{0.5, 0.55, \dots, 0.95\}} AP_{IoU}^{101} \quad (4)$$

where AP is the average AP^{101} over varying IoUs, and AP_{IoU}^{101} is the AP^{101} over the threshold of IoU.

For instance segmentation, the metrics are slightly different from object detection: IoU is calculated by masks instead of bounding boxes. In this study, the metrics in Table 2 were used to evaluate the performance of detectors on the MOCS dataset.

6.2. Detectors for evaluation

Fifteen deep-learning-based detectors were selected to train on and evaluate the MOCS dataset. With the rapid development of deep-learning algorithms, some promising early detectors were not

evaluated in this study. All the detectors evaluated were from after 2015 (Faster R-CNN was proposed, which is the first end-to-end and the first near-real-time deep-learning detector).

Specifically, seven detectors were selected for object detection, and four detectors were selected for instance segmentation. Meanwhile, according to previous research [18], the accuracy of a detector depends heavily on its feature extraction networks, which are referred to as “backbone networks.” Therefore, two detectors with three different backbones were compared. Owing to space constraints, only an overview of the detectors is presented. The details of the detectors can be found in a previous report [18].

6.2.1. Detectors for object detection

Many deep-learning-based object detectors have been proposed in recent years. They can be roughly divided into three types: one-stage detectors, two-stage detectors, and others [39]. To provide a comprehensive evaluation of the MOCS dataset, the most well-known detectors for these types were selected.

For one-stage detectors, YOLO-v3 [7], SSD300 [6], and RetinaNet [40], which are considered milestone detectors and are widely used, were selected. FCOS [41] and NAS-FPN [42], which were proposed in 2019, were also selected. For two-stage detectors, Faster R-CNN [8] has been considered a baseline for many object detection tasks, such as aerial object detection [43] and pedestrian detection [44]. Therefore, Faster R-CNN, with a backbone of ResNet50, was evaluated. In addition to this, TridentFast [39], which is a novel detector designed for a trident block, was also evaluated. The backbones, implementation platforms, mean average precision (mAP) on the COCO dataset, and inference time of these detectors are presented in Table 3.

6.2.2. Detectors for instance segmentation

For instance segmentation, Mask R-CNN [15] is considered one of the most flexible and efficient frameworks for instance segmentation. Therefore, Mask R-CNN, with a backbone of ResNet50, was evaluated. In addition to Mask R-CNN, the recently proposed Mask Scoring RCNN (MS RCNN) [49], SOLO [50], and PointRend [51] were also selected to evaluate the MOCS dataset. The backbones, implementation platforms, mAP of the bounding box and mask, and inference time are presented in Table 4.

6.2.3. Detectors with different backbones

To evaluate the effects of different backbones, Faster-RCNN and

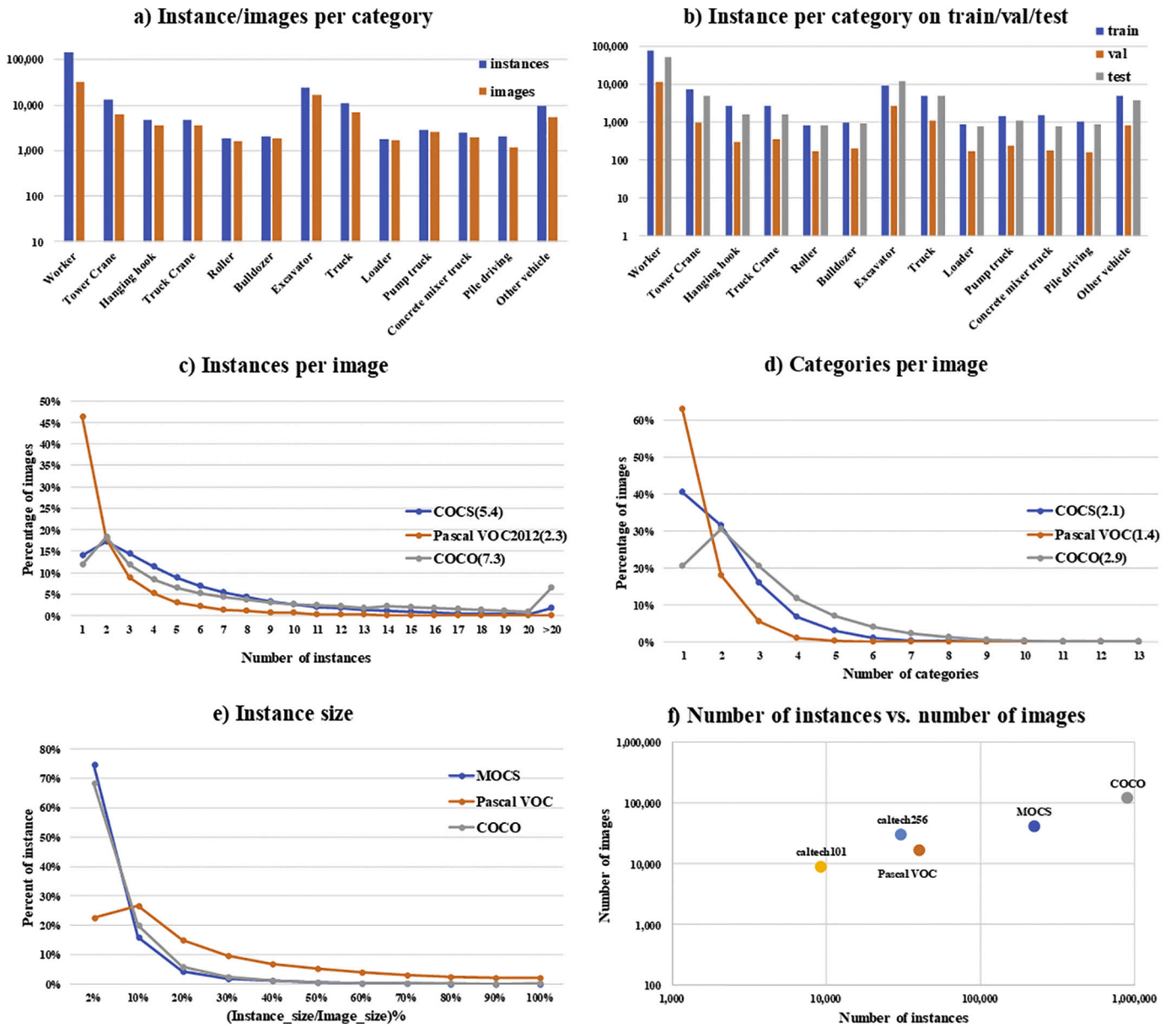


Fig. 11. Statistics of MOCS dataset.

(a) Number of annotated instances/image per category in MOCS, (b) number of annotated instances per category in the split MOCS, (c, d) distribution of images with different numbers of instances/categories for MOCS, Pascal VOC, and COCO2017, (e) distribution of instance size for MOCS, Pascal VOC, and COCO2017, and (f) number of images versus number of instances of popular object recognition datasets.

Table 2

Metrics to evaluate the performance of detectors.

Metrics	Notes
AP	Mean value of AP in each class
$AP^{IoU=0.50}$	AP^{IoU} at IoU = 0.50
$AP^{IoU=0.75}$	AP^{IoU} at IoU = 0.75
AP^{small}	AP for small objects (area of object $< 32^2$ pixels)
AP^{medium}	AP for medium objects (32^2 pixels $<$ area of object $< 96^2$ pixels)
AP^{large}	AP for large objects (area of object $> 96^2$ pixels)
Inference speed	Calculation time on single image
AP for each category	AP for each category

Mask-RCNN with backbones of ResNet50, ResNet101, and ResNeXt101 were selected for training and testing. Details of the selected detectors are listed in Table 5.

6.3. Experimental results

The training settings of all detectors are listed in Table 3. Owing to the limitation of computing resources, the backbones pretrained on the ImageNet were used instead of training anew. All the detectors were trained and tested on an Intel Xeon E5-2637 v4 CPU with two GeForce GTX 1080Ti GPUs. Meanwhile, the optimization method of all these detectors was stochastic gradient descent. Details of the training settings are listed in Table 6. Other settings not mentioned in this study are the same as the original implementations.

Table 3
Detectors for object detection.

Detectors	Backbone	Framework	COCO mAP (bbox)	Inference time (fps)
YOLO-v3 [7]	Darknet53 [45]	Darknet [45]	33.0	30.00 ^a
SSD300 [6]	VGG16 [46]	MMDetection [55]	25.7	25.90 ^b
RetinaNet [40]	ResNet50 [47]	Detection2 [56]	37.9	18.18 ^b
FCOS [41]	ResNet50 [47]+FPN [48]	MMDetection [55]	36.9	11.60 ^b
NAS-FPN [46]	ResNet50 [47]	MMDetection [55]	39.8	13.10 ^b
Faster R-CNN [8]	ResNet50 [47]+FPN [48]	Detection2 [56]	40.2	26.32 ^b
TridentFast [39]	ResNet50 [47]+FPN [48]	Detection2 [56]	40.4	10.30 ^b

^a The inference speed was tested on a Pascal Titan X GPU.

^b The inference speed was tested on a Tesla V100 GPU.

Table 4
Detectors for instance segmentation.

Detectors	Backbone	Framework	COCO mAP (bbox)	COCO mAP (seg)	Inference time (fps)
Mask R-CNN [15]	ResNet50 [47]+FPN [48]	Detection2 [56]	41.0	37.2	23.25 ^a
MS RCNN [49]	RexNet50 [47]+FPN [48]	MMDetection [55]	38.2	35.9	10.10 ^a
SOLO [50]	ResNet50 [47]+FPN [48]	MMDetection [55]	/	35.8	12.98 ^a
PointRend [51]	RexNet50 [47]+FPN [48]	Detection2 [56]	/	38.3	/

^a The inference speed was tested on a Tesla V100 GPU.

Table 5
Detectors with different backbones.

Detectors	Backbone	Framework	COCO mAP (bbox)	COCO mAP (seg)	Inference time (fps)
Faster RCNN [8]	ResNet50 [47]+FPN [48]	Detection2 [56]	40.2	/	26.32 ^a
	ResNet101 [47]+FPN [48]	Detection2 [56]	42.0	/	19.61 ^a
	ResNeXt101 [52]+FPN [48]	Detection2 [56]	43.0	/	10.20 ^a
Mask RCNN [15]	ResNet50 [47]+FPN [48]	Detection2 [56]	41.0	37.2	23.26 ^a
	ResNet101 [47]+FPN [48]	Detection2 [56]	42.9	38.6	17.86 ^a
	RexNeXt101 [52]+FPN [48]	Detection2 [56]	44.3	39.5	9.71 ^a

^a The inference speed was tested on a Tesla V100 GPU.

Table 6
Training settings of detectors.

Backbone	Input size ^a	Lr sched (decay steps) ^b	Initial learning rate ^c
YOLO-v3 (Darknet53)	608 × 608	120e (80e, 100e)	0.0005 ^d
SSD300 (VGG16)	300 × 300	120e (80e, 100e)	0.0005 ^d
RetinaNet (ResNet50 + FPN)	1333 × 800	55e (50e, 40e)	0.005
FCOS (ResNet50 + FPN)	1333 × 800	55e (50e, 40e)	0.005
NAS-FPN (ResNet50)	640 × 640	55e (50e, 40e)	0.02
Faster R-CNN (ResNet50 + FPN)	1333 × 800	55e (50e, 40e)	0.005
Faster R-CNN (ResNet101 + FPN)	1333 × 800	55e (50e, 40e)	0.005
Faster R-CNN (ResNeXt101 + FPN)	1333 × 800	55e (50e, 40e)	0.005
TridentFast (R50)	1333 × 800	55e (50e, 40e)	0.005
Mask RCNN (ResNet50 + FPN)	1333 × 800	55e (50e, 40e)	0.005
Mask RCNN (ResNet101 + FPN)	1333 × 800	55e (50e, 40e)	0.005
Mask RCNN (ResNeXt101 + FPN)	1333 × 800	55e (50e, 40e)	0.005
MS RCNN (ResNet50 + FPN)	1333 × 800	55e (50e, 40e)	0.005
SOLO (ResNet50 + FPN)	1333 × 800	55e (50e, 40e)	0.0025
PointRend (RexNet50 + FPN)	1333 × 800	55e (50e, 40e)	0.005

^a The original input size of detectors was used.

^b Lr sched (decay steps) is the training schedule for detectors. 120e means the detectors have been trained 120 epochs on the whole MOCs train set. (80e, 100e) means the learning rate is reduced to 0.1 × Initial learning rate after 80 epochs and further reduced to 0.01 × Initial learning rate after 100 epochs.

^c Because the batch sizes are different from the baselines, the initial learning rates of the detectors are adjusted according to the linear learning rate scaling rule [53].

^d For YOLO-v3 and SSD, the initial learning rate is reduced to 5e-4 to avoid exploding losses [44].

6.3.1. Results for object detection

After the training, the detectors were evaluated on the test dataset. The overall performance of the test dataset and the inference speed of the detectors can be found in Table 4. For the task of object detection, PointRend with a backbone of ResNet50 achieved the highest mAP (51.039) (Table 7). This is because detectors for segmentation can benefit from multitask learning [15] (i.e., segmentation and bounding box location). For single-task (i.e., bounding box detection) detectors, TridentFast achieved the highest mAP (50.686). Meanwhile, YOLO-v3 had the highest inference speed (27.03 fps), which could push object detection into real-time detection on common devices. In addition to the overall performance of the detectors, the mAP of each category is listed in Table 8. Several detection results in the images are shown in Section 6.3.4.

6.3.2. Results for instance segmentation

The results for instance segmentation are listed in Table 9 and results for each categories are listed in Table 10. PointRend achieved the highest mAP (47.906) with a light backbone of ResNet50 (Table 9). Meanwhile, the inference speed of the MS RCNN with a backbone of ResNet50 was the fastest (9.70 fps). The segmentation results for each category are shown in Table 10. Several segmentation results are shown in the images in Section 6.3.4.

Table 7
Bounding box mAP for object detection.

Backbone	mAP	AP ⁵⁰	AP ⁷⁵	AP ^{small}	AP ^{medium}	AP ^{large}	Inference speed (fps)
YOLO-v3 (Darknet53)	39.05	65.59	41.66	9.60	26.82	51.03	27.03
SSD300 (VGG16)	36.08	59.28	38.03	4.28	19.30	51.20	21.28
RetinaNet (ResNet50 + FPN)	50.01	72.84	53.60	14.77	38.05	62.40	6.80
FCOS (ResNet50 + FPN)	46.85	69.32	50.02	13.05	34.23	60.35	14.93
NAS-FPN (ResNet50)	47.60	68.55	50.06	7.01	30.64	64.03	17.61
Faster R-CNN (ResNet50 + FPN)	50.64	74.65	55.59	19.30	39.59	61.33	8.39
TridentFast (ResNet50)	50.69	73.08	54.71	15.34	37.78	63.63	4.05
Mask R-CNN (ResNet50 + FPN)	50.83	74.89	55.98	19.01	39.63	62.05	7.66
MS RCNN (ResNet50 + FPN)	47.35	70.30	51.79	14.33	35.57	60.70	9.70
SOLO (ResNet50 + FPN)	48.73	67.65	52.02	9.79	32.74	64.97	6.25
PointRend (RexNet50 + FPN)	51.04	74.79	55.79	19.74	39.55	62.13	8.87

Table 8
Bounding box AP for object detection (each category).

Backbone	Worker	Tower Crane	Hanging Head	Crane	Roller	Bulldozer	Excavator	Truck	Loader	Pump Truck	Concrete Mixer	Pile Driver	Other Vehicle
YOLO-v3 (Darknet53)	42.551	38.876	16.782	36.224	48.211	49.694	56.809	40.859	41.126	41.898	44.585	24.943	25.021
SSD300 (VGG16)	30.854	30.945	9.994	31.832	50.996	52.038	56.159	38.333	44.572	43.450	44.790	16.513	18.509
RetinaNet (ResNet50 + FPN)	52.074	46.971	26.167	44.921	62.638	62.913	67.824	50.204	53.721	55.251	57.350	31.852	38.297
FCOS (ResNet50 + FPN)	48.132	44.562	24.350	42.783	59.491	59.620	64.154	46.245	49.516	51.018	55.209	28.542	35.426
NAS-FPN (ResNet50)	45.802	46.090	21.338	43.222	59.437	61.313	67.256	48.609	51.968	58.666	56.214	30.529	28.289
Faster R-CNN (ResNet50 + FPN)	53.653	52.992	33.162	45.622	62.038	60.310	65.945	49.690	51.044	53.711	56.443	32.842	40.854
TridentFast (ResNet50)	52.919	54.267	27.396	45.366	61.426	61.661	67.159	50.364	53.534	54.388	57.216	34.900	38.321
Mask R-CNN (ResNet50 + FPN)	53.891	53.036	32.404	46.125	61.351	60.337	66.128	50.045	51.955	54.107	57.464	33.088	40.897
MS RCNN (RexNet50 + FPN)	50.925	49.330	20.900	43.256	59.793	58.529	64.221	47.059	49.365	50.390	55.617	29.111	37.071
SOLO (ResNet50)	46.758	47.363	17.193	45.160	60.062	64.451	68.882	48.030	54.123	58.958	53.939	33.028	35.592
PointRend (RexNet50 + FPN)	54.499	53.381	32.826	46.152	62.434	60.944	66.590	50.113	52.434	54.078	56.923	32.065	41.064

Table 9
Mask mAP for instance segmentation.

Backbone	mAP	AP ⁵⁰	AP ⁷⁵	AP ^{small}	AP ^{medium}	AP ^{large}	Inference speed (fps)
Mask R-CNN (ResNet50 + FPN)	43.18	71.27	45.35	14.93	32.15	55.05	7.66
MS RCNN (RexNet50 + FPN)	42.86	66.85	45.98	13.50	31.97	56.20	9.70
SOLO (ResNet50 + FPN)	43.64	68.79	45.95	8.18	29.57	59.33	6.25
PointRend (RexNet50 + FPN)	47.91	73.15	52.29	14.98	34.00	61.52	8.87

6.3.3. Results of different backbones

Faster R-CNN and Mask R-CNN with three different backbones were trained and evaluated in this study. For both Faster R-CNN and Mask R-CNN, detectors with a backbone of ResNeXt101 achieved the highest mAP (Table 11). However, heavy backbones have a low inference speed, which limits the real-time performance of detectors on common devices (i.e., 1080ti GPU). Results for each categories are listed in Table 12.

6.3.4. Image results

The image results of detection are shown in this section. Five images

that were never seen in the training process are selected from the test dataset. The object detection results can be seen in Fig. 12, and the segmentation results can be seen in Fig. 13. Instances detected in images are marked with boxes or translucence masks. The colors are randomly selected to distinguish instances. The image in the first column is from the top view, which was captured by the UAV. The second-column image was captured at night. The third-column image contains many categories of objects. The image in the fourth column contains crowded objects, and the image in the fifth column contains many small objects.

Table 10

Mask mAP for instance segmentation (each category).

Backbone	Worker	Tower Crane	Hanging Hook	Vehicle Crane	Roller	Bulldozer	Excavator	Truck	Loader	Pump Truck	Concrete Mixer	Pile Driver	Other Vehicle
Mask R-CNN (ResNet50 + FPN)	51.434	30.771	38.639	30.386	60.421	58.300	53.281	47.387	49.726	25.761	54.204	20.531	40.438
MS RCNN (ResNet50 + FPN)	48.324	34.767	23.606	31.484	60.099	58.677	54.142	46.477	50.150	35.266	54.661	22.153	37.349
SOLO (ResNet50 + FPN)	43.416	35.398	17.589	33.015	59.506	62.740	57.273	48.227	52.494	42.515	53.793	25.356	35.958
PointRend (ResNet50 + FPN)	53.118	42.646	38.622	35.995	64.026	61.534	57.778	49.808	52.616	44.396	56.625	23.310	42.306

Table 11

Results of Faster R-CNN and Mask R-CNN with different backbones.

Detectors	Backbone	mAP	AP ⁵⁰	AP ⁷⁵	AP ^{small}	AP ^{medium}	AP ^{large}	Inference speed
Faster R-CNN	ResNet50	50.64	74.65	55.59	19.30	39.59	61.33	8.39
	ResNet101	50.67	73.98	55.31	18.30	38.84	62.48	6.79
	ResNeXt101	52.45	74.96	57.59	20.14	40.91	64.21	4.01
Mask R-CNN	ResNet50	50.83	74.89	55.98	19.01	39.63	62.05	7.66
	ResNet101	53.39	75.57	58.38	20.23	41.12	65.34	6.22
	ResNeXt101	54.76	76.79	60.07	20.43	43.35	66.57	4.16
Mask R-CNN (seg)	ResNet50	43.18	71.27	45.35	14.93	32.15	55.05	7.66
	ResNet101	45.35	72.74	47.99	15.38	33.64	57.59	6.22
	ResNeXt101	46.88	74.12	49.80	16.17	36.02	58.94	4.16

7. Discussion

7.1. Discussion of MOCS dataset

A growing number of researchers, engineers, and construction companies have begun to use computer vision to conduct progress monitoring, quality analysis, and safety management of the construction site [1–5,14–16,23,24,27–38]. Accurate detection and identification of moving objects by means of digital images/videos form the basis of these tasks [1–5,14–16,23,24]. Although many large-scale datasets for object detection have been released, a dataset for objects at construction sites remains absent, and this greatly limits the development of research in this field. In this study, a large-scale dataset was constructed for detecting and segmenting moving objects at construction sites. The dataset is significant in at least four ways.

- 1) Real site and large scale: Instead of collecting images from the Internet, all the images were collected from 174 real construction sites. This brings the dataset into a line with actual construction scenarios, and the detectors trained on the dataset can be applied well to the site. Moreover, the dataset contains more than 41,668 images and 222,861 instances, which exceeds some large-scale common datasets, such as PASCAL VOC.
- 2) Various categories: Compared with several previous datasets, such as AIM [24], more categories of objects in construction sites are annotated in the MOCS dataset. The dataset includes most of the common moving objects in the construction site, which broadens the application of detectors trained on the dataset.
- 3) Diversity: The MOCS dataset covers a variety of visual conditions in real construction sites, including visual ranges, viewpoints, weather conditions, illumination conditions, individual postures, occlusions, work categories, and construction project categories. The diversity of the MOCS dataset greatly enhances the generalization of the detectors trained on the dataset.
- 4) Precise annotation: A bounding box is used for the object detection task. In addition to the bounding box, the mask of each object is

annotated in the dataset. The mask is mainly used for instance segmentation, which helps to understand the layout of the instance and to analyze the scene [19].

7.2. Discussion of benchmark results

A benchmark was created on the dataset. Seven detectors for object detection, four detectors for instance segmentation, and two detectors with three different backbones were trained and tested on the dataset. The results presented in Section 6.3 demonstrate that, although the detectors differ in speed and precision owing to their architectures, all detectors trained on the MOCS dataset could accurately detect the moving objects at the construction sites. Meanwhile, the detectors trained on MOCS were robust to viewpoints, occlusions, noise on the construction sites, and illumination conditions (Figs. 12 and 13). Tables 8, 10, and 12 show that the mAPs of “Hanging hook” and “Pile driver” are lower than those of the other categories. For “Hanging hook,” the main reason for low mAPs is that the hanging hooks are very small in the images and difficult to detect. For “Pile driver,” the reason is that the shapes of pile drivers are varied, while the images in MOCS are relatively few.

YOLO-v3, Faster R-CNN with ResNet50, and Mask R-CNN with ResNet50 were sampled for comparison with the COCO dataset (Table 13). The mAPs of MOCS are 6–10 points higher than those of COCO. Table 13 shows that the higher mAPs on MOCS mainly benefit from higher mAPs for large objects. However, for medium and small objects, the mAPs of MOCS are lower than those of COCO. Fig. 14 shows several cases of failure detection. The main reason why small and medium mAPs on MOCS are lower than those of COCO is that the construction sites are more chaotic than other scenes, and the features of some unrelated construction components are similar to the small and medium moving objects (Fig. 12a). In addition, occlusions always prevent the detectors from accurately detecting the objects (Fig. 14b), and crowded objects could also confuse the detectors (Fig. 12c). Furthermore because most of the images in training are upright, the detectors had difficulty recognizing objects in reversed images (Fig. 14d).

Table 12
Results of Faster R-CNN and Mask R-CNN with different backbones (each category).

Detectors	Backbone	Worker	Tower Crane	Hanging Head	Crane	Roller	Bulldozer	Excavator	Truck	Loader	Pump Truck	Concrete Mixer	Pile Driver	Other Vehicle
Faster R-CNN	ResNet50	53.653	52.992	33.162	45.622	62.038	60.310	65.945	49.690	51.044	53.711	56.443	32.842	40.854
	ResNet101	53.109	52.737	32.191	45.796	62.397	62.217	65.785	49.001	52.245	53.649	57.188	32.645	39.874
	ResNeXt101	55.458	54.431	33.935	47.046	63.606	62.998	67.381	50.676	54.315	55.707	58.166	35.774	42.365
Mask R-CNN	ResNet50	53.891	53.036	32.404	46.125	61.351	60.337	66.128	50.045	51.955	54.107	57.464	33.088	40.897
	ResNet101	54.958	56.173	33.497	49.398	63.543	64.002	69.190	51.615	54.198	59.199	60.288	35.354	42.623
	ResNeXt101	56.753	56.959	35.505	49.036	65.569	65.005	70.019	53.245	57.591	59.093	60.132	38.639	44.306
Mask R-CNN (seg)	ResNet50	51.434	30.771	38.639	30.386	60.421	58.300	53.281	47.387	49.726	25.761	54.204	20.531	40.438
	ResNet101	52.667	33.257	40.210	33.415	61.732	60.726	55.183	48.699	51.781	30.095	56.614	22.069	42.499
	ResNeXt101	54.063	33.629	41.050	33.715	63.758	62.502	56.875	50.288	54.219	30.883	57.218	25.809	44.257

7.3. Limitations and future work

In this study, a large-scale image dataset, MOCS, was constructed and released. The dataset is intended to help researchers and engineers develop algorithms for detecting moving objects in construction sites and develop more applications of construction automation. Meanwhile, a benchmark was created in this study that provides a guideline for others to select detectors for their applications. However, there are several limitations that should be mentioned.

First, 41,000 construction site images are included in the MOCS dataset, while only 13 categories of objects are labeled. Much information contained in images is not used because of the lack of annotation. For instance, to recognize construction activities from images, it is necessary to detect moving objects and structural elements (installed formwork, reinforcement bar, scaffolding, etc.) [16]. Such annotations are not available in the current MOCS dataset. Another case is excavator safety monitoring, which sometimes requires more-detailed annotations, such as key points of the excavator [36]. The current MOCS dataset lacks such annotations. In addition, detectors trained on the current MOCS could not detect workers not using hardhats [14] because such information is not labeled. Therefore, annotations in MOCS should be greatly enriched to develop more-advanced construction applications. The dataset is publicly available, and any improvement in the MOCS dataset is welcomed.

Second, although 41,000 images are collected in MOCS, instances are relatively few for several categories (e.g., pile drivers), and some construction machine categories (e.g., road pavers and tunnel boring machines) are not included in the current MOCS dataset. Meanwhile, images are collected in limited types of construction site in China and Pakistan, and detectors trained on the MOCS dataset may not be applicable to other sites in other countries. Therefore, for a specific project, it is strongly recommended that detectors be trained on the MOCS dataset first, and then transfer learning strategies, such as fine tuning, should be used to obtain a better result. In the future, it is planned to extend the MOCS dataset to include more categories of objects and more types of site worldwide. It is also hoped that researchers can extend the MOCS dataset or develop their own datasets to verify each other.

Third, the MOCS dataset is an image dataset and currently is focused on object detection. To develop more computer vision applications in the field of construction, more types of dataset (e.g., video datasets and 3D point cloud datasets) should be developed. For instance, to analyze the activity of earthmoving equipment, a dataset containing videos of earthmoving operation must be developed [5]. Another example is that, to develop algorithms for progress monitoring or building information model creation, datasets of 3D point clouds are sometimes required. Such datasets are scarce the construction research community. Continuing work in this area is planned so that more datasets can be made publicly available.

Apart from the MOCS datasets, a benchmark for the dataset is also provided. Researchers and engineers can select and develop detectors based on the benchmark results. However, the detectors were implemented and trained according to the original implementations. For better results, the following techniques can be applied. First, the detectors were trained without data augmentation in this research. Data augmentation methods, such as image flipping, rotation, and clipping, can be used to obtain better results [54]. Second, for current detectors, it is difficult to detect small objects. To improve the performance of small object detection, several parameters (e.g., anchor box scale for Faster R-CNN) could be tuned. Third, the detectors presented in this research were trained on two GPUs. In practice, more GPUs and larger batch sizes can be used to achieve higher accuracy.

8. Conclusions

A large-scale image dataset, MOCS, was introduced. The purpose of the dataset is to train and test computer vision algorithms for detecting

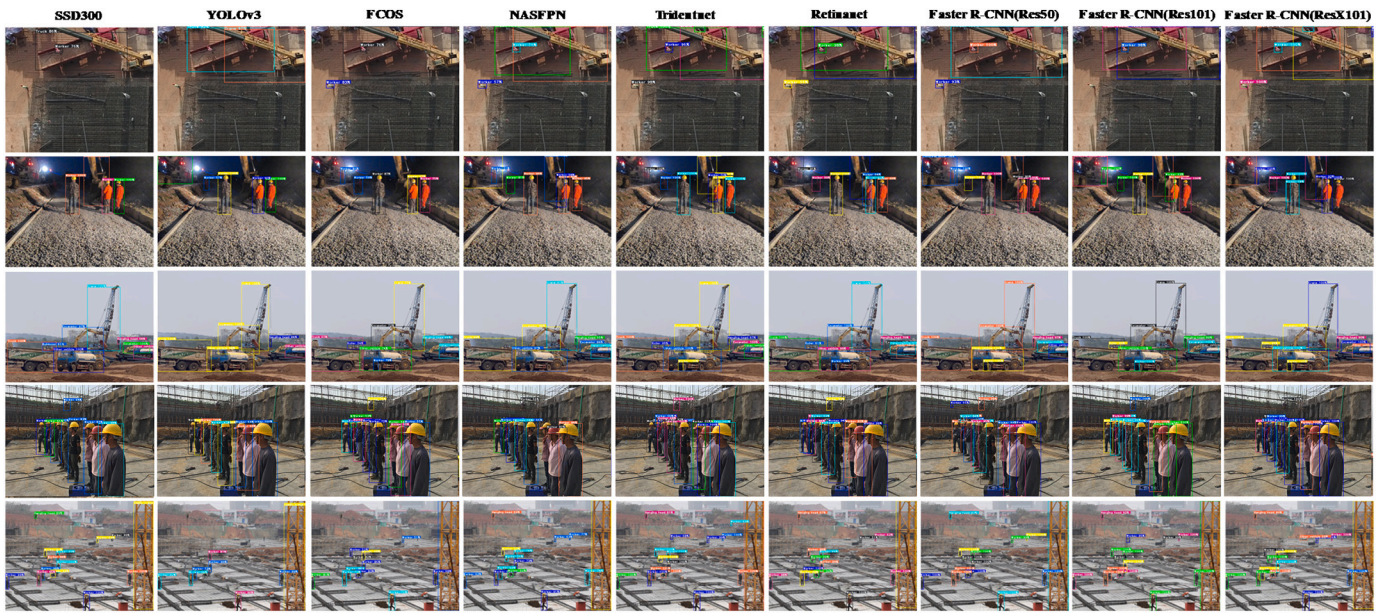


Fig. 12. Image results of object detection.

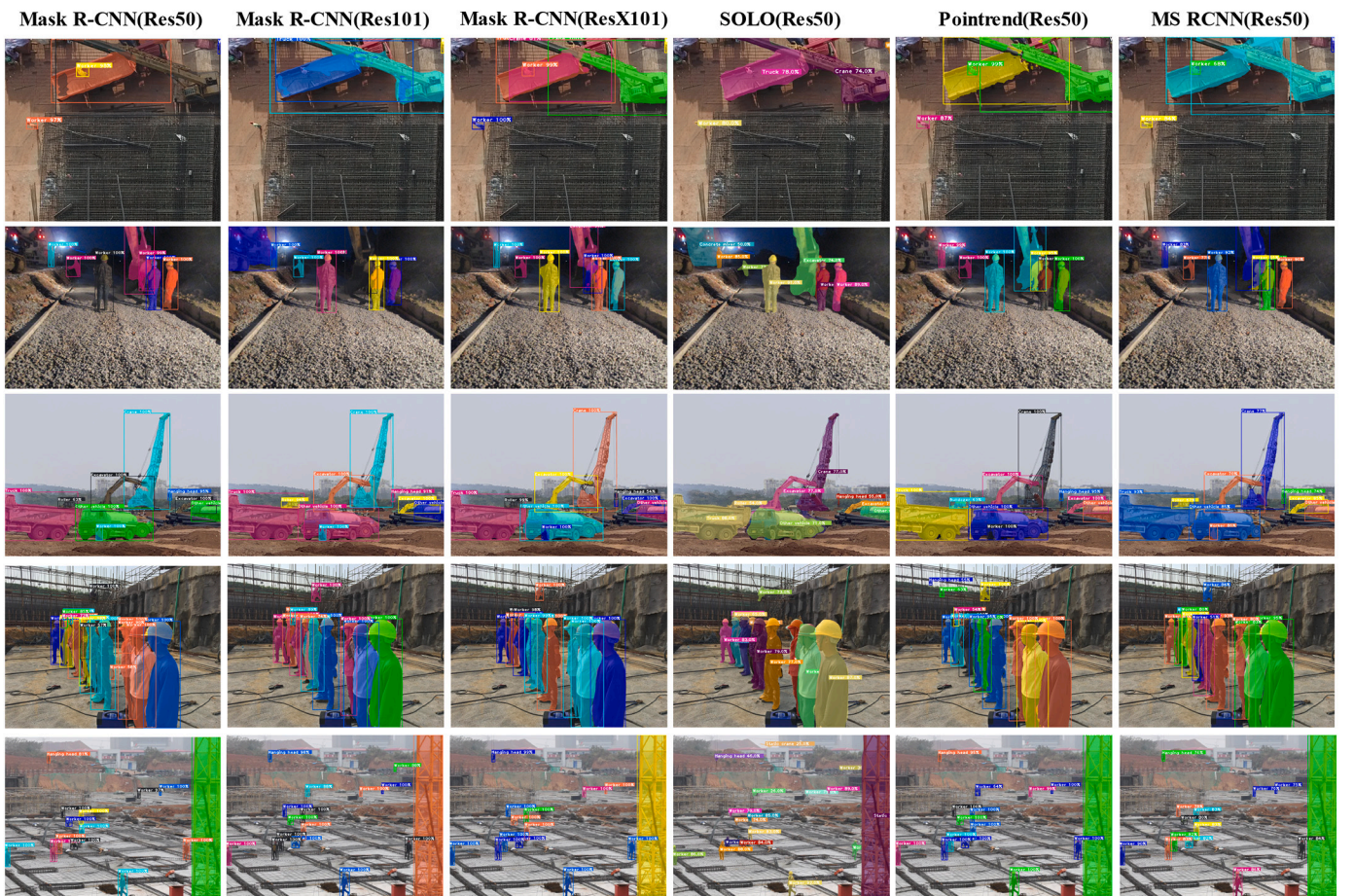


Fig. 13. Image results of instance segmentation.

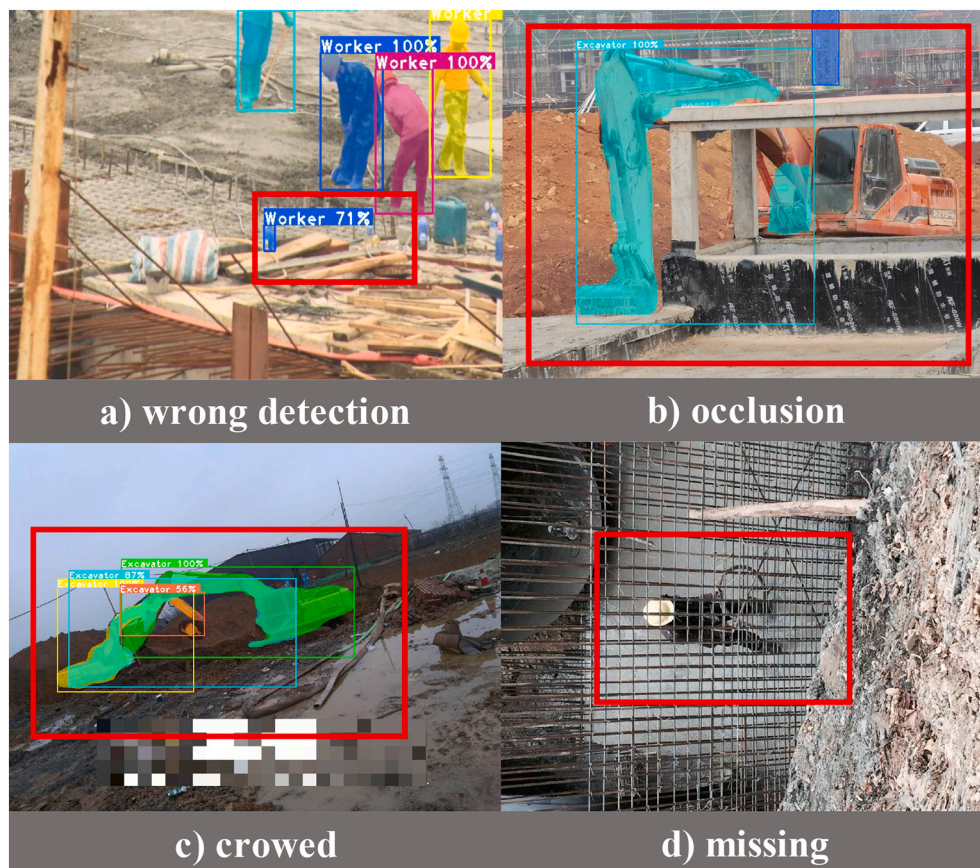
objects at construction sites. For the dataset, 41,668 images were collected from real construction sites, and more than 222,861 instances of 13 categories were precisely annotated with the styles of both bounding box and mask. The dataset is publicly available, and it is

believed that releasing such a dataset can greatly benefit researchers in this field. Meanwhile, a benchmark was created on the dataset. Seven detectors for object detection, four detectors for instance segmentation, and two detectors with three different backbones were evaluated. It is

Table 13

Results of comparison with COCO dataset.

Detectors	Dataset	mAP	AP ⁵⁰	AP ⁷⁵	AP ^{small}	AP ^{medium}	AP ^{large}
YOLO-v3	MOCS	39.0	65.6	41.7	9.6	26.8	51.0
	COCO	33.0	57.9	34.4	18.3	35.4	41.9
Faster R-CNN ResNet50	MOCS	50.6	74.6	55.6	19.3	39.6	61.3
	COCO	40.2	61.0	43.8	24.2	43.5	52.0
Mask R-CNN ResNet50	MOCS ^{bbox}	50.8	74.9	56.0	19.0	39.6	62.0
	COCO ^{bbox}	41.0	61.5	44.9	24.9	43.9	53.3
	MOCS ^{segm}	43.2	71.3	45.4	14.9	32.2	55.1
	COCs ^{segm}	37.2	58.6	39.9	18.6	39.5	53.3

**Fig. 14.** Failure cases of Mask R-CNN with ResXNet101.

hoped that this can provide a reference for others. Several challenges and future directions for image datasets and improvements for detecting objects in construction sites were discussed.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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References

- [1] J. Gong, C.H. Caldas, An object recognition, tracking, and contextual reasoning-based video interpretation method for rapid productivity analysis of construction operations, *Autom. Constr.* 20 (8) (2011) 1211–1226, <https://doi.org/10.1016/j.autcon.2011.05.005>.
- [2] M. Golparvar-Fard, F. Peña-Mora, C.A. Arboleda, S. Lee, Visualization of construction progress monitoring with 4D simulation model overlaid on time-lapsed photographs, *J. Comput. Civ. Eng.* 23 (6) (2009) 391–404, [https://doi.org/10.1061/\(asce\)0887-3801\(2009\)23:6\(391\)](https://doi.org/10.1061/(asce)0887-3801(2009)23:6(391)).
- [3] M.W. Park, N. Elsafty, Z. Zhu, Hardhat-wearing detection for enhancing on-site safety of construction workers, *J. Constr. Eng. Manag.* 141 (9) (2015) 04015024, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000974](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000974).
- [4] D. Kim, M. Liu, S. Lee, V.R. Kamat, Remote proximity monitoring between mobile construction resources using camera-mounted UAVs, *Autom. Constr.* 99 (2019) 168–182, <https://doi.org/10.1016/j.autcon.2018.12.014>.
- [5] D. Roberts, M. Golparvar-Fard, End-to-end vision-based detection, tracking and activity analysis of earthmoving equipment filmed at ground level, *Autom. Constr.* 105 (2019) 102811, <https://doi.org/10.1016/j.autcon.2019.04.006>.

- [6] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.Y. Fu, A.C. Berg, Ssd: single shot multibox detector, in: European Conference on Computer Vision, Springer, Cham, 2016, October, pp. 21–37, https://doi.org/10.1007/978-3-319-46448-0_2.
- [7] J. Redmon, A. Farhadi, YoloV3: An Incremental Improvement, arXiv preprint arXiv: 1804.02767, <https://arxiv.org/abs/1804.02767>, 2018.
- [8] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: towards real-time object detection with region proposal networks, in: Advances in Neural Information Processing Systems, 2015, pp. 91–99, <https://doi.org/10.1109/TPAMI.2016.2577031>.
- [9] L. Fei-Fei, R. Fergus, P. Perona, One-shot learning of object categories, IEEE Trans. Pattern Anal. Mach. Intell. 28 (4) (2006) 594–611, <https://doi.org/10.1109/TPAMI.2006.79>.
- [10] Griffin, G., Holub, A., & Perona, P., Caltech-256 Object Category Dataset, <http://authors.library.caltech.edu/7694/>, 2007, CaltechAUTHORS:CNS-TR-2007-001.
- [11] T.Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, C.L. Zitnick, Microsoft coco: Common objects in context, in: European conference on computer vision, Springer, Cham, 2014, September, pp. 740–755, https://doi.org/10.1007/978-3-319-10602-1_48.
- [12] Everingham, M., Eslami, S. A., Van Gool, L., Williams, C. K., Winn, J., & Zisserman, A. (2015). The pascal visual object classes challenge: a retrospective. Int. J. Comput. Vis., 111(1), pp.98–136. doi:<https://doi.org/10.1007/s11263-014-0733-5>.
- [13] M. Everingham, L. Van Gool, C.K. Williams, J. Winn, A. Zisserman, The pascal visual object classes (voc) challenge, Int. J. Comput. Vis. 88 (2) (2010) 303–338, <https://doi.org/10.1007/s11263-009-0275-4>.
- [14] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T.M. Rose, W. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, Autom. Constr. 85 (2018) 1–9, <https://doi.org/10.1016/j.autcon.2017.09.018>.
- [15] Q. Fang, H. Li, X. Luo, L. Ding, T.M. Rose, W. An, Y. Yu, A deep learning-based method for detecting non-certified work on construction sites, Adv. Eng. Inform. 35 (2018) 56–68, <https://doi.org/10.1016/j.aei.2018.01.001>.
- [16] X. Luo, H. Li, D. Cao, F. Dai, J. Seo, S. Lee, Recognizing diverse construction activities in site images via relevance networks of construction-related objects detected by convolutional neural networks, J. Comput. Civ. Eng. 32 (3) (2018) 04018012, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000756](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000756).
- [17] K. He, G. Gkioxari, P. Dollár, R. Girshick, Mask r-cnn, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2961–2969, <https://doi.org/10.1109/ICCV.2017.322>.
- [18] Z. Zou, Z. Shi, Y. Guo, J. Ye, Object Detection in 20 Years: A Survey, arXiv preprint arXiv:1905.05055, <https://arxiv.org/abs/1905.05055v2>, 2019.
- [19] B. Hariharan, P. Arbeláez, R. Girshick, J. Malik, Simultaneous detection and segmentation, in: European Conference on Computer Vision, Springer, Cham, 2014, September, pp. 297–312, https://doi.org/10.1007/978-3-319-10584-0_20.
- [20] N. Dalal, B. Triggs, Histograms of oriented gradients for human detection, in: 2005 IEEE computer society conference on computer vision and pattern recognition (CVPR '05), Vol. 1, IEEE, 2005, June, pp. 886–893, <https://doi.org/10.1109/cvpr.2005.177>.
- [21] P. Felzenszwalb, D. McAllester, D. Ramanan, A discriminatively trained, multiscale, deformable part model, in: 2008 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, 2008, June, pp. 1–8, <https://doi.org/10.1109/CVPR.2008.4587597>.
- [22] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, Y. LeCun, Overfeat: Integrated Recognition, Localization and Detection using Convolutional Networks, arXiv preprint arXiv:1312.6229, <https://arxiv.org/abs/1312.6229>, 2013.
- [23] D. Roberts, T. Bretl, M. Golparvar-Fard, Detecting and classifying cranes using camera-equipped UAVs for monitoring crane-related safety hazards, in: Computing in Civil Engineering 2017, 2017, pp. 442–449, <https://doi.org/10.1061/9780784480847.055>.
- [24] H. Kim, H. Kim, Y.W. Hong, H. Byun, Detecting construction equipment using a region-based fully convolutional network and transfer learning, J. Comput. Civ. Eng. 32 (2) (2018) 04017082, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000731](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000731).
- [25] Z. Zhang, S. Fidler, R. Urtasun, Instance-level segmentation for autonomous driving with deep densely connected mrfs, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 669–677, https://doi.org/10.1007/978-3-319-46466-4_19.
- [26] A. Arnab, P.H. Torr, Bottom-up Instance Segmentation using Deep Higher-order Crfs, arXiv preprint arXiv:1609.02583, <https://arxiv.org/abs/1609.02583>, 2016.
- [27] V. Balali, M. Golparvar-Fard, Segmentation and recognition of roadway assets from car-mounted camera video streams using a scalable non-parametric image parsing method, Autom. Constr. 49 (2015) 27–39, <https://doi.org/10.1016/j.autcon.2014.09.007>.
- [28] G.A. Atkinson, W. Zhang, M.F. Hansen, M.L. Holloway, A.A. Napier, Image segmentation of underfloor scenes using a mask regions convolutional neural network with two-stage transfer learning, Autom. Constr. 113 (2020) 103118, <https://doi.org/10.1016/j.autcon.2020.103118>.
- [29] F. Wei, G. Yao, Y. Yang, Y. Sun, Instance-level recognition and quantification for concrete surface bughole based on deep learning, Autom. Constr. 107 (2019) 102920, <https://doi.org/10.1016/j.autcon.2019.102920>.
- [30] J. Wu, N. Cai, W. Chen, H. Wang, G. Wang, Automatic detection of hardhats worn by construction personnel: a deep learning approach and benchmark dataset, Autom. Constr. 106 (2019) 102894, <https://doi.org/10.1016/j.autcon.2019.102894>.
- [31] S. Takahashi, S.I. Kaneko, Motion tracking of crane hook based on optical flow and orientation code matching, in: 2008 10th IEEE International Workshop on Advanced Motion Control, IEEE, 2008, March, pp. 149–152, <https://doi.org/10.1109/AMC.2008.4516057>.
- [32] Y. Li, S. Wang, B. Li, Improved visual hook capturing and tracking for precision hoisting of tower crane, Adv. Mech. Eng. 5 (2013) 426810, <https://doi.org/10.1155/2013/426810>.
- [33] Y. Guo, Y. Xu, S. Li, Dense construction vehicle detection based on orientation-aware feature fusion convolutional neural network, Autom. Constr. 112 (2020) 103124, <https://doi.org/10.1016/j.autcon.2020.103124>.
- [34] T. Slaton, C. Hernandez, R. Akhavan, Construction activity recognition with convolutional recurrent networks, Autom. Constr. 113 (2020) 103138, <https://doi.org/10.1016/j.autcon.2020.103138>.
- [35] H. Tajee, Z. Zhu, Image dataset development for measuring construction equipment recognition performance, Autom. Constr. 48 (2014) 1–10, <https://doi.org/10.1016/j.autcon.2014.07.006>.
- [36] H. Luo, M. Wang, P.K. Wong, J.C.P. Cheng, Full body pose estimation of construction equipment using computer vision and deep learning techniques, Autom. Constr. 110 (2020) 103016, <https://doi.org/10.1016/j.autcon.2019.103016>.
- [37] X. Huang, T. Sasaki, H. Hashimoto, F. Inoue, B. Zheng, T. Masuda, K. Ikeuchi, An accurate and efficient pile driver positioning system using laser range finder, in: International Workshop on Depth Image Analysis and Applications, Springer, Berlin, Heidelberg, 2013, pp. 158–167, https://doi.org/10.1007/978-3-642-40303-3_17.
- [38] T. Sasaki, H. Kawahara, F. Inoue, H. Hashimoto, Circle fitting based pile positioning and machine pose estimation from range data for pile driver navigation, IFAC Proc. Vols. 45 (22) (2012) 848–853, <https://doi.org/10.3182/20120905-3-HR-2030.00137>.
- [39] Y. Li, Y. Chen, N. Wang, Z. Zhang, Scale-aware trident networks for object detection, in: In Proceedings of the IEEE International Conference on Computer Vision, 2019, pp. 6054–6063, <https://doi.org/10.1109/ICCV.2019.00615>.
- [40] T.Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 2980–2988, <https://doi.org/10.1109/TPAMI.2018.2858826>.
- [41] Z. Tian, C. Shen, H. Chen, T. He, Fcos: Fully convolutional one-stage object detection, in: Proceedings of the IEEE International Conference on Computer Vision, 2019, pp. 9627–9636, <https://doi.org/10.1109/ICCV.2019.00972>.
- [42] G. Ghiasi, T.Y. Lin, Q.V. Le, Nas-fpn: Learning scalable feature pyramid architecture for object detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2019, pp. 7036–7045, <https://doi.org/10.1109/CVPR.2019.00720>.
- [43] G.S. Xia, X. Bai, J. Ding, Z. Zhu, S. Belongie, J. Luo, L. Zhang, DOTA: A large-scale dataset for object detection in aerial images, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 3974–3983, <https://doi.org/10.1109/CVPR.2018.00418>.
- [44] M. Braun, S. Krebs, F. Flohr, D.M. Gavrilu, The Eurocity Persons Dataset: A Novel Benchmark for Object Detection. <https://arxiv.org/abs/1805.07193>, 2018.
- [45] J. Redmon, Darknet: Open Source Neural Networks in C. <https://pjreddie.com/darknet/>, 2013. Accessed date: July 4, 2020.
- [46] K. Simonyan, A. Zisserman, Very Deep Convolutional Networks for Large-Scale Image Recognition. <https://arxiv.org/abs/1409.1556>, 2014.
- [47] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778, <https://doi.org/10.1109/CVPR.2016.90>.
- [48] T.Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature pyramid networks for object detection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2117–2125, <https://doi.org/10.1109/CVPR.2017.106>.
- [49] Z. Huang, L. Huang, Y. Gong, C. Huang, X. Wang, Mask scoring r-cnn, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2019, pp. 6409–6418, <https://doi.org/10.1109/CVPR.2019.00657>.
- [50] X. Wang, T. Kong, C. Shen, Y. Jiang, L. Li, SOLO: Segmenting Objects by Locations. <https://arxiv.org/pdf/1912.04488>, 2019. Accessed date: July 20 2020.
- [51] A. Kirillov, Y. Wu, K. He, R. Girshick, PointRend: Image Segmentation as Rendering. <https://arxiv.org/abs/1912.08193>, 2019. Accessed date: June 18 2020.
- [52] S. Xie, R. Girshick, P. Dollár, Z. Tu, K. He, Aggregated residual transformations for deep neural networks, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 1492–1500, <https://doi.org/10.1109/CVPR.2017.634>.
- [53] P. Goyal, P. Dollár, R. Girshick, P. Noordhuis, L. Wesolowski, A. Kyrola, K. He, Accurate, Large Minibatch Sgd: Training Imagenet in 1 Hour. <https://arxiv.org/abs/1706.02677>, 2017.
- [54] Z. Zhang, T. He, H. Zhang, Z. Zhang, J. Xie, M. Li, Bag of Freebies for Training Object Detection Neural Networks. <https://arxiv.org/abs/1902.04103>, 2019.
- [55] K. Chen, J. Wang, J. Pang, Y. Cao, Y. Xiong, X. Li, Z. Zhang, MMDetection: Open Mmlab Detection Toolbox and Benchmark. <https://arxiv.org/abs/1906.07155>, 2019.
- [56] Y. Wu, A. Kirillov, F. Massa, W.Y. Lo, R. Girshick, Detectron2. <https://github.com/facebookresearch/detectron2>, 2019.
- [57] G. Bradski, The OpenCV Library. Dr. Dobb's Journal of Software Tools. <https://opencv.org/>, 2000.